



VP AIR TEST NEET (2025-26)

PART TEST - 09

DROPPER

DURATION: 180 Minutes

Date: 01-02-2026

M. MARKS: 720

Topics Covered

Physics:	Electromagnetic Induction, Alternating Current, Electromagnetic Waves
Chemistry:	Aldehydes, Ketones and Carboxylic Acids, Amines, Bio-molecules, Purification and Analysis of Organic Compound
Botany:	Respiration in Plants, Photosynthesis in Higher Plants
Zoology:	Reproductive Health, Evolution

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

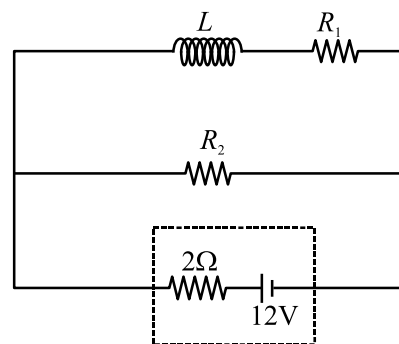
Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

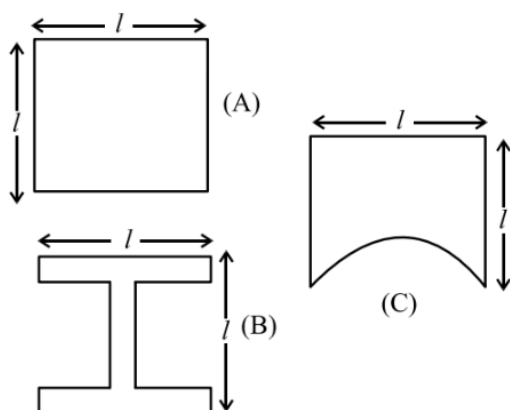
Student's Signature: _____ **Invigilator's Signature:** _____

- The magnetic flux linked with a coil (in Weber) is given by the equation $\phi = 5t^2 + 3t + 16$. The magnitude of the induced EMF in the coil at $t = 4$ seconds is:
 (1) 33 V (2) 43 V
 (3) 10 V (4) 23 V
- An electromagnetic wave traveling in the negative x -direction has a magnetic field vector given by $\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{j}$ Tesla. What is the maximum electric field and its direction?
 (1) 60 V/m along $-x$ -axis
 (2) 60 V/m along $-z$ -axis
 (3) 30 V/m along $+y$ -axis
 (4) 60 V/m along $+z$ -axis
- In a series LCR circuit connected to an AC source $V = 200 \sin(100t)$, the components are $R = 80 \Omega$, $L = 0.4$ H, and $C = 100 \mu\text{F}$. The impedance of the circuit is:
 (1) 80Ω (2) 100Ω
 (3) 141Ω (4) 120Ω
- The current in a solenoid of 40 turns, having a length of 4π cm and area 12 cm^2 , changes at the rate of 1 As^{-1} . Find the emf induced in it.
 (1) $1.92 \times 10^{-4} \text{ V}$
 (2) $1.92 \times 10^{-5} \text{ V}$
 (3) $1.92 \times 10^{-6} \text{ V}$
 (4) $1.92 \times 10^{-3} \text{ V}$
- A coil of resistance 10Ω and inductance 5 H is connected to a galvanometer. The magnetic flux through the coil changes from 25 Wb to 5 Wb in 2 seconds. The total charge flown through the coil is:
 (1) 1 C (2) 2 C
 (3) 10 C (4) 5 C
- An inductor of inductance $L = 400$ mH and resistors of $R_1 = 2 \Omega$ and $R_2 = 2 \Omega$ are connected to a battery of EMF 12 V as shown in figure. The internal resistance of the battery is 2Ω . At $t = 0$ (switch closed), the current drawn from the battery is:

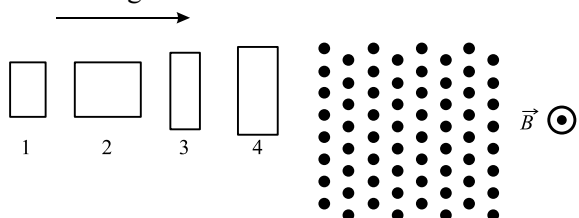


- (1) 3 A (2) 1.5 A
 (3) 6 A (4) 2 A
- A transformer with 80% efficiency is used to step down the voltage from 220 V to 22 V . If the current flowing in the secondary circuit is 2 A , what is the current flowing in the primary coil?
 (1) 0.20 A (2) 0.25 A
 (3) 0.40 A (4) 20 A
- In an AC circuit, voltage $V = 100 \sin(100t)$ volt and current $I = 100 \sin(100t + \pi/3)$ mA. The average power dissipated in the circuit is:
 (1) 2.5 W (2) 5.0 W
 (3) 10.0 W (4) 2500 W
- Two coils have a mutual inductance of 0.05 H . The current changes in the first coil according to $I = 10 \sin(100\pi t)$. The maximum value of EMF induced in the second coil is:
 (1) $50\pi \text{ V}$ (2) $5\pi \text{ V}$
 (3) $\pi \text{ V}$ (4) $2\pi \text{ V}$
- A circular loop is placed in a uniform magnetic field of 0.04 T with its plane perpendicular to the field. The radius of the loop starts shrinking at a constant rate. At an instant when the radius is 5 cm , the induced EMF in the loop is found to be $4\pi \mu\text{V}$ (microvolts). What is the rate at which the radius is decreasing?
 (1) 1 m/s (2) 1 cm/s
 (3) 1 mm/s (4) 2 mm/s
- A capacitor of capacitance C is charged to potential V and discharged through resistance R . The displacement current at time t is given by:
 (1) $\frac{V}{R} e^{-t/RC}$ (2) $\frac{V}{R}$
 (3) $\frac{V}{R} (1 - e^{-t/RC})$ (4) Zero

12. Various metallic sheets are exposed to changing magnetic field perpendicular to their planes and eddy currents are produced in them. The minimum value of eddy current will be observed in:



- (1) A
(2) B
(3) C
(4) Same in all
13. Two coaxial solenoids are arranged with a short solenoid placed inside a long solenoid. The outer solenoid has a length of 50 cm and 4000 turns. The inner solenoid has a length of 10 cm, 100 turns, and a cross-sectional area of 10 cm^2 . The mutual inductance of this system is:
(Take $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$)
- (1) $16\pi \times 10^{-5} \text{ H}$
(2) $32\pi \times 10^{-5} \text{ H}$
(3) $32\pi \times 10^{-4} \text{ H}$
(4) $160\pi \times 10^{-5} \text{ H}$
14. Four conducting rectangular loops are moving with the same constant velocity $\vec{v}$ towards a region of uniform magnetic field $\vec{B}$. The vertical edge lengths (perpendicular to the velocity) of loops 1, 2, 3, and 4 are L , L , $2L$, and $3L$ respectively as shown in figure.



Rank the loops according to the maximum magnitude of the induced EMF (ϵ) as they enter the magnetic field.

- (1) $\epsilon_4 > \epsilon_3 > \epsilon_2 > \epsilon_1$
(2) $\epsilon_4 > \epsilon_3 > \epsilon_2 = \epsilon_1$
(3) $\epsilon_2 > \epsilon_4 > \epsilon_3 > \epsilon_1$
(4) $\epsilon_1 = \epsilon_2 = \epsilon_3 = \epsilon_4$

15. An LR circuit contains an inductor of 10 H and a resistor of 10Ω connected to a 20 V battery. How long does it take for the current to reach half its maximum value? ($\ln 2 = 0.693$.)

- (1) 0.693 s (2) 1.386 s
(3) 0.5 s (4) 2.0 s

16. The magnetic energy stored in a solenoid of length 1 m and cross-section area 10^{-3} m^2 having 1000 turns carrying 10 A current is:

- (1) $0.02 \pi \text{ J}$ (2) $2 \pi \text{ J}$
(3) 0.02 J (4) 10 J

17. A rectangular loop is pulled out of a uniform magnetic field with a constant velocity v . The power required to pull the loop is proportional to:

- (1) v (2) v^2
(3) v^3 (4) Independent of v

18. The self-inductance of a coil is 5 mH. Current changes from 1 A to 4 A in 0.05 s. The magnitude of average induced EMF is:

- (1) 3 V (2) 30 V
(3) 0.03 V (4) 0.3 V

19. A choke coil is preferred over a resistor in AC circuits to reduce current because:

- (1) Choke coils generate more heat.
(2) Choke coils reduce current with minimal power consumption.
(3) Choke coils increase the power factor.
(4) Choke coils convert AC to DC.

20. A copper disc of radius a rotates with frequency f in a uniform magnetic field B perpendicular to the disc. Two brushes are connected to the center and the rim. The potential difference generated is:

- (1) $\pi B a^2 f$ (2) $2\pi B a^2 f$
(3) $B a^2 f$ (4) $\frac{1}{2} B a^2 f$

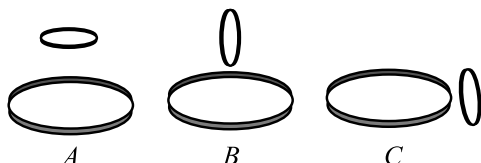
21. In a series RL circuit connected to an AC source, the phase difference between voltage and current is 45° . If the frequency of the source is doubled, the new phase difference will be: (keeping other parameter remains constant)

- (1) $\tan^{-1}(2)$ (2) 45°
(3) 90° (4) $\tan^{-1}(1/2)$

22. The resonant frequency of a series LCR circuit is ν_0 . If the capacitance is made 4 times the initial value, what should be the change in inductance to keep the resonant frequency same?

- (1) Increase by 4 times
- (2) Decrease to 1/4th of original
- (3) Increase by 2 times
- (4) Decrease to 1/2 of original

23. Two circular coils can be arranged in any of the three situation shown in the figure. Their mutual inductance will be



- (1) Maximum in situation (A)
- (2) Maximum in situation (B)
- (3) Maximum in situation (C)
- (4) The same in all situations

24. A plane electromagnetic wave travels in free space along the positive x-axis. At a particular point, $\vec{E} = 6.3\hat{j}$ V/m. The corresponding magnetic field $\vec{B}$ is:

- (1) $2.1 \times 10^{-8} \hat{k}$ T
- (2) $2.1 \times 10^{-8} \hat{j}$ T
- (3) $2.1 \times 10^{-8} (-\hat{k})$ T
- (4) $18.9 \times 10^{-8} \hat{k}$ T

25. An AC voltage source $V = 200\sqrt{2}\sin(100t)$ volt is connected across a pure capacitor of $100\mu\text{F}$. The RMS current in the circuit is:

- (1) 1 A
- (2) 2 A
- (3) 4 A
- (4) 0.5 A

26. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): In a series LCR circuit connected to an AC source, at resonance, the current is maximum.

Reason (R): At resonance, the inductive reactance cancels the capacitive reactance, making the circuit purely resistive with maximum impedance.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

27. Given below are two statements:

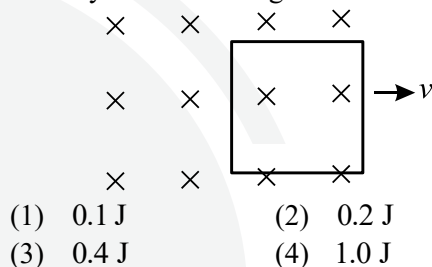
Statement I: Modified Ampere's Law states $\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d)$, where I_d is displacement current.

Statement II: Displacement current is produced only when the electric field is constant.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

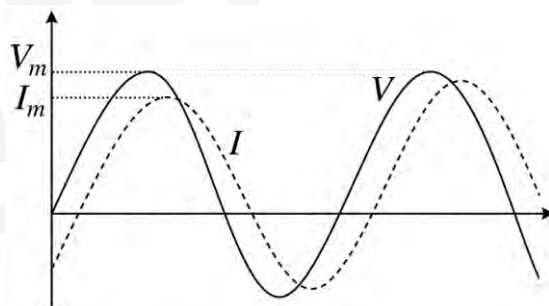
- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

28. A square loop of side 1 m and resistance 5Ω is placed in a uniform magnetic field of 2 T perpendicular to the plane of the loop. If the loop is pulled out of the field uniformly in 2 s, the work done by the external agent is:



- (1) 0.1 J
- (2) 0.2 J
- (3) 0.4 J
- (4) 1.0 J

29. The given graph shows the variation of instantaneous emf (V) and current (I) with time in an AC circuit. The circuit element(s) present in the circuit is/are:



- (1) C-R series circuit.
- (2) Only a pure capacitor (C)
- (3) Only a pure inductor (L)
- (4) L-R series circuit

30. A plane electromagnetic wave is propagating in vacuum along the direction $\hat{d} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$. If the electric field at an instant is $\vec{E} = E_0 \hat{k}$, then the direction of the magnetic field $\vec{B}$ is:

- (1) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$
- (2) $\frac{1}{\sqrt{2}}(\hat{i} - \hat{j})$
- (3) $\frac{1}{\sqrt{2}}(-\hat{i} + \hat{j})$
- (4) $-\hat{k}$

31. An EM wave travels in a non-magnetic medium with relative permittivity $\epsilon_r = 9$. If the peak electric field in this medium is 12 V/m, the peak magnetic field B_0 is:

- (1) 4×10^{-8} T (2) 1.2×10^{-7} T
(3) 3.6×10^{-8} T (4) 2.4×10^{-7} T

32. In an EM wave, the amplitude of the magnetic field is B_0 . The total average energy density of the wave is:

- (1) $\frac{B_0^2}{2\mu_0}$ (2) $\frac{B_0^2}{\mu_0}$
(3) $\frac{B_0^2}{4\mu_0}$ (4) $\frac{B_0}{2\mu_0^2}$

33. The electric field of an EM wave is $\vec{E} = 40 \cos(kz - 6 \times 10^8 t) \hat{i}$ V/m. If the wave is traveling in a medium with refractive index $n = 2$, the value of the wave vector k is:

- (1) 2 m^{-1} (2) 4 m^{-1}
(3) 1 m^{-1} (4) 0.5 m^{-1}

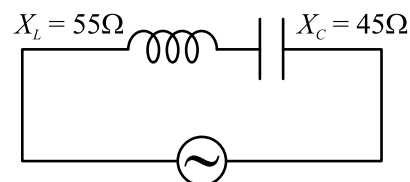
34. Match **List-I** (Electromagnetic Wave Type) with **List-II** (Typical Application/Field) and select the **correct** option from the choices given below:

List-I (EM Wave)		List-II (Application)	
(A)	Ultraviolet Rays	(I)	Therapeutic treatment of muscular strain
(B)	Microwaves	(II)	Precision ophthalmic surgery (LASIK)
(C)	Infrared Radiation	(III)	Avionic navigation and Radar systems
(D)	X-rays	(IV)	Diagnostic imaging of skeletal fractures

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-III, C-II, D-I

35. In given A.C. circuit shown, the RMS value of current is:



$$V = 20 \sin(100 \pi t) \text{ volt}$$

- (1) 2 A (2) 0.5 A
(3) $\sqrt{2}$ A (4) $\left(\frac{1}{\sqrt{2}}\right)$ A

36. In a plane electromagnetic wave travelling in vacuum the electric field oscillates sinusoidally at a frequency of 30×10^{10} Hz and amplitude 96 V m^{-1} .

Choose the **correct** statements.

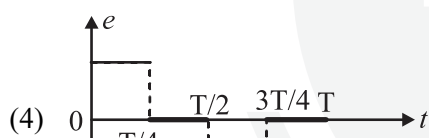
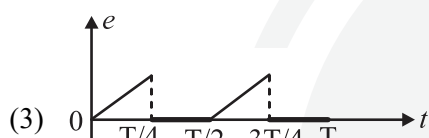
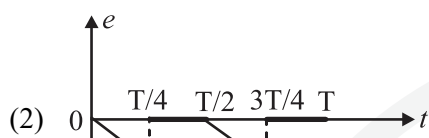
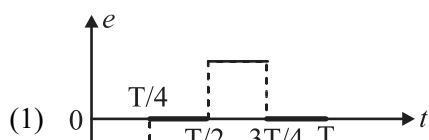
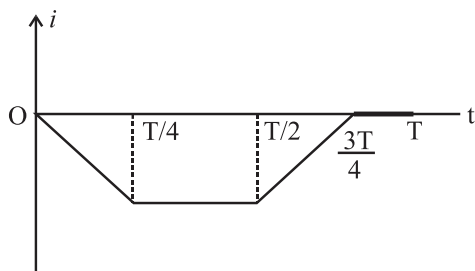
- A. The wavelength of the wave is 0.02 m
B. Amplitude of the oscillating magnetic field is $32 \times 10^{-8} \text{ T}$
C. The phase difference between the electric field and magnetic field is zero.

- (1) A and B only
(2) B and C only
(3) A and C only
(4) A, B and C

37. The electric field component of an electromagnetic wave is given by $E = 60 \sin \omega \left(t - \frac{x}{c} \right) \text{ N/C}$ (where t is in second and x is in meter). The total energy contained in a cylinder of cross-section 10 cm^2 and length $\frac{1}{8.85} \text{ cm}$ along the x -axis will be: (symbols have their usual meaning)

- (1) $3.2 \times 10^{-14} \text{ J}$
(2) $6.2 \times 10^{-14} \text{ J}$
(3) $2.8 \times 10^{-14} \text{ J}$
(4) $1.8 \times 10^{-14} \text{ J}$

38. The current i in a coil varies with time t as shown in the figure. The variation of induced emf e with time t is represented by:



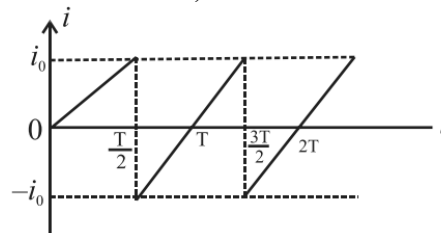
39. Two identical resistors are heated for the same amount of time by two different currents. The first resistor carries a direct current (DC) of 10 A. The second resistor carries a sinusoidal alternating current (AC) with a peak value of 10 A. What will be the ratio of the heat produced in the first resistor (by DC) to the heat produced in the second resistor (by AC)?

- (1) 1 : 1 (2) 1 : 2
(3) 2 : 1 (4) 4 : 1

40. An inductor of 0.4 H and a resistor of $30\ \Omega$ are connected in series to an AC source. The source provides an RMS voltage of 100 V and has an angular frequency of 100 rad/s. What is the wattless component of the RMS current flowing in the circuit?

- (1) 1.2 A (2) 1.6 A
(3) 2.0 A (4) 2.5 A

41. The current i in an AC circuit varies with time t as shown in the figure. What is the root mean square (RMS) value of the current for one complete cycle (from $t = 0$ to $t = T$)?



- (1) $\frac{i_0}{\sqrt{3}}$ (2) $\frac{i_0}{\sqrt{2}}$
(3) $\frac{i_0}{2}$ (4) 0

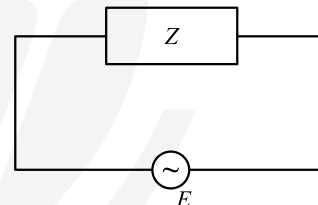
42. Consider a parallel plate capacitor connected across a battery. If net electric flux in the capacitor at time t is $\phi = \frac{3t^2}{\epsilon_0}$ (V-m) where t is in second, then

displacement current between its plates at $t = 5$ sec is;

- (1) 5 A (2) 20 A
(3) 30 A (4) 50 A

43. In a box Z of unknown elements, an ac voltage $E = E_0 \sin(\omega t + \phi)$ is applied and current in the circuit was found to be $I = I_0 \sin\left(\omega t + \phi + \frac{\pi}{4}\right)$.

The unknown element in the box may be:



- (1) Only capacitor
(2) Inductor and resistor both connected in series
(3) Either (capacitor, resistor and inductor) connected in series or (capacitor and resistor) connected in series
(4) Only resistor

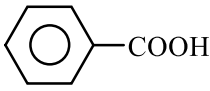
44. To minimize eddy current losses in a transformer, the core is typically:

- (1) Made of a single solid piece of highly permeable magnetic material.
(2) Constructed from thick sheets of steel insulated from each other.
(3) Made of several thin laminated sheets of soft iron, electrically insulated from each other.
(4) Both (1) and (2)

45. A conducting circular loop is placed in a uniform magnetic field, $B = 0.025$ T with its plane perpendicular to the direction of field. The radius of the loop is made to shrink at constant rate of 3 mm s^{-1} . The induced EMF when the radius is 2 cm, is:

- (1) $2\ \mu\text{V}$ (2) $2\pi\ \mu\text{V}$
(3) $\pi\ \mu\text{V}$ (4) $3\pi\ \mu\text{V}$

46. Weakest acid among the following is:

- (1) $\text{CH}_3\text{-COOH}$
- (2) HCOOH
- (3) 
- (4) ClCH_2COOH

47. Hinsberg reagent is:

- (1) $\text{CH}_3\text{-}$  $\text{-SO}_2\text{Cl}$
- (2)  $\text{-SO}_2\text{Cl}$
- (3)  -COCl
- (4) $\text{C}_2\text{H}_5\text{-}$  $\text{-SO}_2\text{Cl}$

48. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Glycogen is a monosaccharide.

Reason R: Glycogen is not sweet in taste.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both A and R are true, and R is the correct explanation of A.
- (2) Both A and R are true, but R is not the correct explanation of A.
- (3) A is true, but R is false.
- (4) A is false, but R is true.

49. Given below are two statements:

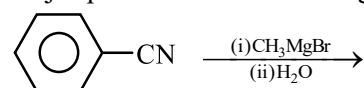
Statement I: Thin layer chromatography is based on the principle of differential adsorption.

Statement II: Partition chromatography is based on continuous differential partitioning of components of a mixture between stationary and mobile phases.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

50. Major product of the following reaction is:



- (1) Mesityloxide
- (2) Acetophenone
- (3) Benzophenone
- (4) Benzoquinone

51. The **correct** order of basic strength in vapour phase for methyl amines is:

- (1) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$
- (2) $(\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2$
- (3) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2$
- (4) $(\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH}$

52. Which among the following is an iodinated derivative of amino acid tyrosine?

- (1) Testosterone
- (2) Estradiol
- (3) Thyroxine
- (4) Progesterone

53. Commonly used adsorbent in column chromatography is:

- (1) Zeolite
- (2) Charcoal
- (3) Graphite
- (4) Alumina

54. Given below are two statements:

Statement I: Camphor is well known for its odour.

Statement II: Higher fatty acids are used for manufacture of soaps and detergents.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

55. On heating ethylamine with CHCl_3 and ethanolic KOH, the product obtained is:

- (1) CH_3NC
- (2) $\text{C}_2\text{H}_5\text{CN}$
- (3) $\text{C}_2\text{H}_5\text{NC}$
- (4) CH_3CN

56. Which of the following is **not** found in DNA?

- (1) Uracil
- (2) Adenine
- (3) Guanine
- (4) Cytosine

57. Match List-I with List-II.

List-I		List-II	
(A)	$[\text{Fe}(\text{SCN})]^{2+}$	(I)	Prussian blue
(B)	$[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$	(II)	Black
(C)	$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$	(III)	Violet
(D)	PbS	(IV)	Red

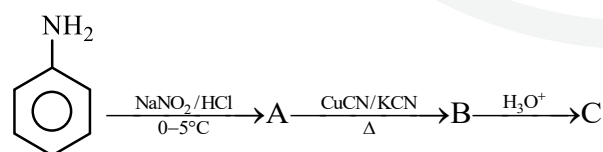
Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-I, C-III, D-II
- (3) A-IV, B-III, C-I, D-II
- (4) A-IV, B-I, C-II, D-III

58. Keto-enol tautomerism is shown by

- (1)
- (2)
- (3)
- (4)

59. The major product (C) obtained in the given reaction sequence is:



- (1)
- (2)
- (3)
- (4)

60. Match List-I with List-II.

List-I (Name of vitamins)		List-II (Deficiency diseases)	
(A)	Vitamin C	(I)	Xerophthalmia
(B)	Vitamin A	(II)	Convulsions
(C)	Vitamin B ₆	(III)	Increased fragility of RBCs
(D)	Vitamin E	(IV)	Scurvy

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-I, C-II, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-III, B-IV, C-II, D-I

61. Given below are two statements:

Statement I: Fractional distillation method is used to separate different fractions of crude oil in petroleum industry.

Statement II: Simple distillation method is used to separate the liquids having sufficient difference in their boiling points.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

62. Which of the following carbonyl compound will undergo aldol condensation reaction with dil. NaOH?

- (1)
- (2) CH_3COCH_3
- (3) HCHO
- (4)

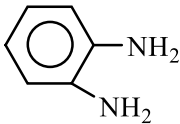
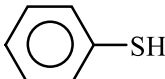
63. Given below are two statements:

Statement I: Coupling reaction is an example of electrophilic substitution reaction.

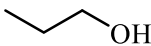
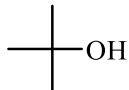
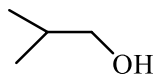
Statement II: $\text{CH}_3\text{NHC}_2\text{H}_5$ is an example of simple amine.

In the light of the above statements, choose the most appropriate answer from the options given below:

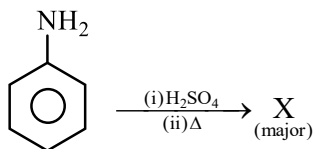
- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

64. Sugar present in milk is known as:
 (1) Lactose (2) Maltose
 (3) Fructose (4) Galactose
65. Which of the following compounds will give blood red colour during lassaighe's test for Nitrogen?
 (1) 
 (2) 
 (3) 
 (4) 
66. Which of the following compounds will **not** give iodoform on reaction with NaOI?
 (1) Isopropyl alcohol
 (2) Acetone
 (3) Isobutyraldehyde
 (4) Ethyl alcohol
67. Molecular formula $C_4H_{11}N$ can form:
 (A) Chain Isomers
 (B) Position Isomers
 (C) Functional Isomers
 (D) Metamers Isomers
 (1) A, B and C only
 (2) B, C and D only
 (3) A, B and D only
 (4) A, B, C and D
68. Bromine water oxidises glucose into:
 (1) Malonic acid (2) Tartaric acid
 (3) Saccharic acid (4) Gluconic acid
69. For organic compounds, Leibig's method is used for the estimation of:
 (1) Carbon and Hydrogen
 (2) Hydrogen and Oxygen
 (3) Carbon and Oxygen
 (4) Carbon and Halogen
70. The major product of the following reaction is:

$$CH_3 - CN \xrightarrow[(ii) H_2O]{(i) AlH(i-Bu)_2}$$

 (1) CH_3CHO (2) $CH_3CH_2NH_2$
 (3) CH_3COOH (4) CH_3CH_2OH
71. Benzonitrile on reduction with $LiAlH_4$ gives:
 (1) Aniline (2) Benzamide
 (3) Benzylamine (4) Benzoquinone
72. Nucleotides are joined together by phosphodiester linkage between which carbon atoms of the pentose sugar?
 (1) 5' and 3' (2) 5' and 1'
 (3) 4' and 1' (4) 4' and 2'
73. 0.2 g of an organic compound containing phosphorus gives 0.111 g $Mg_2P_2O_7$ on analysis. The percentage of phosphorus in the organic compound is:
 (1) 9.2% (2) 15.5%
 (3) 18.9% (4) 11.6%
74. Ethanoic acid on heating with P_2O_5 gives:
 (1) $CH_2=CH_2$ (2) CH_3CHO
 (3) $CH_3COOC_2H_5$ (4) $(CH_3CO)_2O$
75. Number of resonating structure of aniline and anilinium ion respectively are:
 (1) 5 and 5
 (2) 5 and 6
 (3) 5 and 2
 (4) 6 and 5
76. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: During denaturation of proteins they loses their biological activity.
Reason R: During denaturation, globules unfold and helix get uncoiled.
 In the light of the above statements, choose the **correct** answer from the options given below.
 (1) Both A and R are true, and R is the correct explanation of A.
 (2) Both A and R are true, but R is not the correct explanation of A.
 (3) A is true, but R is false.
 (4) A is false, but R is true.
77. Which of the following mixture can be separated into pure compounds by sublimation?
 (1) Naphthalene and anthraquinone
 (2) Anthraquinone and NH_4Cl
 (3) Naphthalene and camphor
 (4) Camphor and $NaCl$
78. Which of the following compounds on oxidation give propanone?
 (1) 
 (2) 
 (3) 
 (4) 

79. Identify (X) in the given reaction:



- (1) p-aminophenol
(2) Benzene sulphonic acid
(3) Sulphanilic acid
(4) Phenyl hydroxyl amine
80. α and β forms of D-glucopyranose are known as
(1) Anomers
(2) Enantiomers
(3) Tautomers
(4) Epimers
81. For the estimation of nitrogen, 0.5 g of an organic compound is treated as per Kjeldahl's method and evolved ammonia is absorbed in 40 ml of $\frac{M}{10}$ H_2SO_4 . The unreacted acid requires 10 ml of $\frac{M}{10}$ NaOH for complete neutralisation. The percentage of nitrogen in organic compound is:
(1) 19.6%
(2) 17.2%
(3) 21.4%
(4) 15.2%
82. Least reactive compound towards nucleophilic addition reaction among the following is:
(1) Benzaldehyde
(2) p-Tolualdehyde
(3) p-Nitrobenzaldehyde
(4) p-Chlorobenzaldehyde
83. Which among the following amine can be prepared by Gabriel phthalimide synthesis?
(1) $\text{C}_6\text{H}_5\text{NH}_2$
(2) CH_3NH_2
(3) $(\text{C}_2\text{H}_5)_3\text{N}$
(4) $\text{CH}_3\text{NHC}_2\text{H}_5$
84. Given below are two statements:
Statement I: Glycine is optically inactive amino acid.
Statement II: Glutamine is an essential amino acid.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

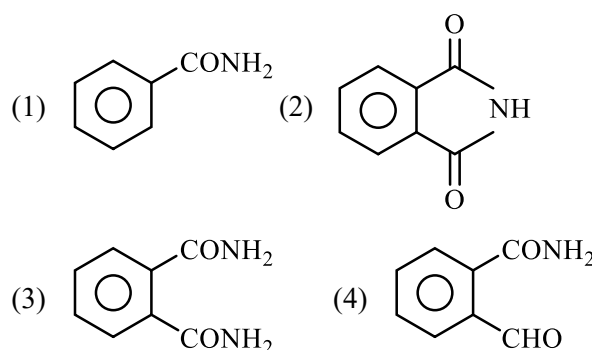
85. The purification method which is based on the principle of 'solubility in two different solvents' is:
(1) Sublimation
(2) Thin layer Chromatography
(3) Partition
(4) Differential extraction

86. Match List-I with List-II.

List-I (Carbonyl derivative)		List-II (Name)	
(A)	$>\text{C}=\text{NR}$	(I)	Phenylhydrazone
(B)	$>\text{C}=\text{N}-\text{OH}$	(II)	Schiff's base
(C)	$>\text{C}=\text{N}-\text{NH}-\text{C}_6\text{H}_5$	(III)	Semicarbazone
(D)	$>\text{C}=\text{N}-\text{NHCONH}_2$	(IV)	Oxime

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
(2) A-IV, B-I, C-II, D-III
(3) A-IV, B-III, C-I, D-II
(4) A-III, B-IV, C-II, D-I
87. During the nitration of aniline in strong acidic medium, the amount of ortho derivative obtained among its nitro derivative at 288 K is:
(1) 0% (2) 47%
(3) 51% (4) 2%
88. **Incorrect** statement regarding Sucrose is:
(1) It is a reducing sugar.
(2) It is dextrorotatory.
(3) It is a disaccharide.
(4) On hydrolysis it produces glucose and fructose.
89. In sodium fusion test of organic compounds, the nitrogen of the organic compound is converted into
(1) NaCN
(2) NaNH_2
(3) NaNO_2
(4) NaNC
90. Phthalamide on strongly heating gives:



91. The complex shown below belongs to mitochondrial ETS. It is the;



- (1) complex I. (2) complex II.
(3) complex III. (4) complex IV.
92. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The ATP synthase enzyme in chloroplast consists of two parts, CF_0 and CF_1 .
Reason (R): The CF_0 component of ATP synthase catalyses ATP synthesis.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.
93. Identify the **correct** sequence of intermediates as they appear in the glycolytic pathway.
(1) Glucose-6-phosphate $\rightarrow$ 3-phosphoglyceraldehyde $\rightarrow$ Fructose-1, 6-bisphosphate $\rightarrow$ Pyruvic acid
(2) Glucose-6-phosphate $\rightarrow$ Fructose-1, 6-bisphosphate $\rightarrow$ 3-phosphoglyceraldehyde $\rightarrow$ Pyruvic acid
(3) Fructose-1, 6-bisphosphate $\rightarrow$ Glucose-6-phosphate $\rightarrow$ 3-phosphoglyceraldehyde $\rightarrow$ Pyruvic acid
(4) Glucose-6-phosphate $\rightarrow$ 3-phosphoglyceraldehyde $\rightarrow$ Pyruvic acid $\rightarrow$ Fructose-1, 6-bisphosphate
94. Select the **correctly** matched pair regarding photosynthesis.
(1) Melvin Calvin - Used ^{14}C to discover the C_4 pathway
(2) Jan Ingenhousz - Showed that only the green parts of plants could release oxygen
(3) Julius von Sachs - Located chlorophyll in special bodies called stomata
(4) T.W. Engelmann - Used aerobic bacteria to detect sites of CO_2 evolution

95. How many of the following statements are **correct** for the process of aerobic respiration?
A. It leads to a partial breakdown of organic substances.
B. Carbon dioxide and water are released.
C. Ethanol and lactic acid are produced as byproducts.
D. Its processes occur in cytoplasm and mitochondria.
E. A large amount of energy is released.
Choose the **correct** option.
(1) One only
(2) Two only
(3) Three only
(4) Four only

96. Match **List-I** with **List-II**.

List-I (Pigment)		List-II (Colour in chromatogram)	
(A)	Carotenoids	(I)	Yellow
(B)	Xanthophylls	(II)	Yellow green
(C)	Chlorophyll b	(III)	Yellow to yellow-orange
(D)	Chlorophyll a	(IV)	Bright or blue green

Choose the *most appropriate* answer from the options given below.

- (1) A-IV, B-III, C-II, D-I
(2) A-IV, B-II, C-III, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-I, C-II, D-IV
97. In the electron transport system, ubiquinone receives reducing equivalents from;
(1) Citrate synthase
(2) NADH dehydrogenase
(3) ATP synthase
(4) Pyruvate dehydrogenase
98. A biologist notes that two distinct perspectives arose from the study of living organisms over time. The perspective focusing on the organismic and higher levels of organization led to the development of:
(1) Ecology and related disciplines.
(2) Physiology and Biochemistry.
(3) Molecular Biology.
(4) Cytology.

99. Identify the **incorrect** statement about the Krebs' cycle.
- (1) The cycle starts with the condensation of acetyl group with OAA and water to yield citric acid.
 - (2) During the conversion of succinyl-CoA to succinic acid, a molecule of GTP is synthesised.
 - (3) There are two points in the cycle where FAD^+ is reduced to FADH_2 .
 - (4) The continued oxidation of acetyl CoA requires the continued replenishment of oxaloacetic acid.

100. What is the primary function of the accessory pigments like chlorophyll *b*, xanthophylls, and carotenoids in photosynthesis?
- (1) To directly convert light energy to chemical energy.
 - (2) To absorb light and transfer the energy to the reaction centre.
 - (3) To form the reaction centre of the photosystems.
 - (4) To absorb only in the blue and red regions of the spectrum.

101. Given below are two statements:

Statement I: During fermentation, more than 13 per cent of the energy in glucose is released.

Statement II: Yeasts poison themselves to death when the concentration of alcohol reaches about 13 per cent.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
 - (2) Both Statement I and Statement II are incorrect.
 - (3) Statement I is correct but Statement II is incorrect.
 - (4) Statement I is incorrect but Statement II is correct.
102. The primary CO_2 acceptor in C_3 and C_4 plants, respectively, are;
- (1) RuBP; PEP
 - (2) PEP; RuBP
 - (3) PGA; OAA
 - (4) OAA; PGA

103. Select the **incorrect** statement regarding the assumptions made when calculating the net gain of ATP in the respiratory balance sheet.
- (1) There is a sequential, orderly pathway functioning.
 - (2) The NADH synthesised in glycolysis is transferred into the mitochondria.
 - (3) Intermediates in the pathway can be withdrawn to synthesise other compounds.
 - (4) Only glucose is being respired and no other alternative substrates enter the pathway.

104. Which of the following is **true** for cyclic photophosphorylation?
- (1) It involves both PS I and PS II.
 - (2) It results in the synthesis of both ATP and $\text{NADPH} + \text{H}^+$.
 - (3) The excited electron from P700 returns to it after passing through an electron transport chain.
 - (4) It occurs in the granal lamellae.

105. During respiration, the strategy of the plant cell is to catabolise glucose in a way that:
- (1) all the liberated energy goes out as heat.
 - (2) the oxidation occurs in a single step to maximize energy release.
 - (3) the energy released can be coupled to ATP synthesis.
 - (4) only anaerobic breakdown occurs to conserve energy.

106. Read the following statements about photorespiration.
- A. Due to this process, the productivity of plants decrease.
 - B. In this pathway, RuBP is converted to 2 molecules of PGA.
 - C. It results in the synthesis of ATP and NADPH.
 - D. It occurs in all plants.
 - E. It is a wasteful process.
- The **correct** statements are;
- (1) B and C
 - (2) A and E
 - (3) B and D
 - (4) C and D

107. Which of the following pairs has compounds with same number of carbon atoms?
- (1) Phosphoglycolate and Acetyl CoA
 - (2) Phosphoglycerate and Oxaloacetate
 - (3) Succinate and α -ketoglutarate
 - (4) Aspartic acid and Tripalmitin

108. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Plants have stomata and lenticels for gaseous exchange but lack specialized respiratory organs.

Reason (R): Each living cell in a plant is located quite close to the surface of the plant, and there is very little transport of gases from one plant part to another.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (3) (A) is true but (R) is false.
 - (4) (A) is false but (R) is true.
109. During chemiosmosis in chloroplasts:
- (1) protons accumulate in the stroma.
 - (2) ATP is synthesised.
 - (3) pH of the thylakoid lumen increases.
 - (4) atmospheric CO_2 is reduced.
110. The respiratory pathway is:
- (1) a purely catabolic process involving only breakdown of substrates.
 - (2) a purely anabolic process involving only synthesis of intermediate compounds.
 - (3) involved in both the catabolism and anabolism of substrates.
 - (4) involved in only breaking down carbohydrates and synthesizing proteins.

111. Match List-I with List-II.

List-I		List-II	
(A)	One turn of glycolysis	(I)	Utilises 1 molecule of H_2O
(B)	One turn of Tricarboxylic acid cycle	(II)	Utilises 2 NADPH
(C)	One turn of Calvin cycle	(III)	Releases 2 molecules of CO_2
(D)	One turn of non-cyclic photophosphorylation	(IV)	Produces 4 ATP

Choose the *most appropriate* answer from the options given below.

- (1) A-IV, B-III, C-II, D-I
- (2) A-IV, B-II, C-III, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-I, C-II, D-IV

112. The regeneration of _____ is crucial for the Calvin cycle to continue uninterrupted.

- (1) CO_2
- (2) O_2
- (3) RuBP
- (4) triose phosphate

113. Arrange the following intermediates of the Krebs' cycle in the sequence they occur after the formation of citric acid.

- A. α -ketoglutaric acid
- B. Succinyl-CoA
- C. Malate
- D. Isocitrate

Choose the **correct** option.

- (1) $\text{D} \rightarrow \text{A} \rightarrow \text{B} \rightarrow \text{C}$
- (2) $\text{A} \rightarrow \text{D} \rightarrow \text{B} \rightarrow \text{C}$
- (3) $\text{D} \rightarrow \text{B} \rightarrow \text{A} \rightarrow \text{C}$
- (4) $\text{A} \rightarrow \text{B} \rightarrow \text{C} \rightarrow \text{D}$

114. Which of the following characteristics is/are **not** true about C_4 plants?

- A. They show CO_2 saturation beyond $450 \mu\text{L}^{-1}$.
- B. They have a special type of leaf anatomy called 'Kranz' anatomy.
- C. C_3 cycle occurs in bundle sheath cells.
- D. The first CO_2 fixation product is a 3-carbon acid.
- E. They tolerate higher temperatures than C_3 plants.

Choose the **correct** option.

- (1) B and C
- (2) C and E
- (3) B only
- (4) A and D

115. During glycolysis, ATP is consumed during the conversion of;

- (1) Glucose to Glucose-6-phosphate
- (2) 1,3-bisphosphoglycerate to 3-phosphoglycerate
- (3) PEP to Pyruvic acid
- (4) Sucrose to glucose and fructose

116. Given below are two statements:

Statement I: Photosynthesis is a physico-chemical process by which green plants use light energy to drive the synthesis of organic compounds.

Statement II: Ultimately, all living forms on earth depend on sunlight for energy.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are incorrect.
- (2) Both Statement I and Statement II are correct.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

117. A compound when used as respiratory substrate in aerobic respiration consumed 10 O₂ molecules and produced 7 CO₂ molecules. Most probably, the compound is a/an;

- (1) carbohydrate.
- (2) protein.
- (3) fatty acid.
- (4) organic acid.

118. The carrier molecule that transports electrons from cytochrome B6f to PS I during light reaction of photosynthesis is;

- (1) plastoquinone.
- (2) plastocyanin.
- (3) ferredoxin.
- (4) ubiquinone.

119. How many molecules of ATP are produced from the complete oxidation of one molecule of NADH in the Electron Transport System?

- (1) Only 1
- (2) Only 2
- (3) Only 3
- (4) Only 4

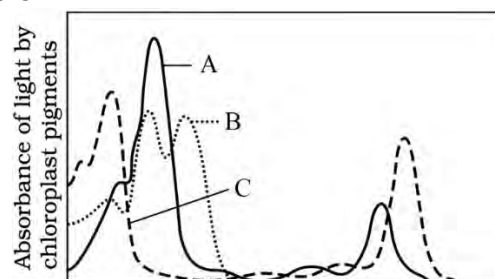
120. The two photosystems, PS I and PS II, are named in the;

- (1) sequence of their discovery.
- (2) sequence of their function in light reaction.
- (3) context of number of reaction centres in them.
- (4) context of number of wavelengths of light absorbed by them.

121. Fermentation takes place under anaerobic conditions in;

- (1) all prokaryotes.
- (2) only unicellular eukaryotes.
- (3) germinating seeds.
- (4) only multicellular eukaryotes.

122. The graph below shows the absorption spectrum of chlorophyll *a*, *b* and the carotenoids. Which of the labels represent the accessory photosynthetic pigments?



- (1) B and C
- (2) A and C
- (3) A, B and C
- (4) A and B

123. Select the **incorrect** pair regarding the location of processes in cellular respiration.

- (1) EMP pathway - Cytoplasm
- (2) Oxidative decarboxylation of pyruvate - Mitochondrial matrix
- (3) Alcoholic fermentation - Mitochondrial matrix
- (4) Electron Transport System - Inner mitochondrial membrane

124. At high light intensities, the rate of photosynthesis does not show further increase because:

- (1) light becomes the limiting factor.
- (2) saturation occurs at 50% of full sunlight.
- (3) the breakdown of chlorophyll occurs.
- (4) other factors like CO₂ concentration or temperature become limiting.

125. Match **List-I** with **List-II** w.r.t the entry of different substrates into the respiratory pathway.

List-I (Substrate)		List-II (Entry point)	
(A)	Carbohydrates	(I)	Acetyl CoA
(B)	Fatty acids	(II)	Pyruvate or Acetyl CoA
(C)	Proteins	(III)	Glucose
(D)	Glycerol	(IV)	PGAL

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
- (2) A-III, B-II, C-I, D-IV
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-I, C-IV, D-II

- 126.** Read the following statements regarding the factors affecting photosynthesis.
- The light reactions are not affected by temperature.
 - Light is rarely a limiting factor in nature for plants in dense forests.
 - The dark reactions are temperature controlled.
 - C₄ plants are not affected by surrounding temperature.
 - Water stress reduces CO₂ availability by causing stomatal closure.
- The **correct** statements are;
- D and E
 - B and D
 - C and E
 - A and C
- 127.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
- Assertion (A):** The pyruvate dehydrogenase enzyme performs oxidative decarboxylation of pyruvate.
- Reason (R):** The pyruvate dehydrogenase enzyme requires several coenzymes for catalysing the reaction.
- In the light of the above statements, choose the *most appropriate* answer from the options given below.
- Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (A) is true but (R) is false.
 - (A) is false but (R) is true.
- 128.** In a photosynthesizing plant, if light becomes unavailable, the biosynthetic process will:
- stop immediately.
 - continue for a long time.
 - continue for some time and then stop.
 - become faster.
- 129.** Which of the following is **not** a substrate level phosphorylation?
- Conversion of Ubiquinol to Ubiquinone
 - Conversion of Succinyl-CoA to Succinic acid
 - Conversion of Triose bisphosphate to Triose phosphate
 - Conversion of Phosphoenol pyruvate to Pyruvate
- 130.** How many of the following are direct products of the light reactions of photosynthesis?
- ATP
 - NADPH
 - O₂
 - Glucose
 - CO₂
- Choose the **correct** option.
- Two only
 - Three only
 - Four only
 - Five
- 131.** For each ATP produced, ____ pass(es) through F₀ from the intermembrane space to the mitochondrial matrix down the electrochemical proton gradient.
- 1 H⁺
 - 2 H⁺
 - 3 H⁺
 - 4 H⁺
- 132.** How many NADH + H⁺ molecules are formed during complete aerobic respiration of one glucose molecule?
- 7
 - 10
 - 6
 - 12
- 133.** The most abundant enzyme in the world is;
- PEPcase.
 - Citrate synthase.
 - RuBisCO.
 - Invertase.
- 134.** Which complex of the mitochondrial ETS is also called the cytochrome c oxidase?
- Complex I
 - Complex II
 - Complex III
 - Complex IV
- 135.** Who amongst the following was **not** involved in proposing the scheme of glycolysis?
- Bassham
 - Meyerhof
 - Parnas
 - Embden

136. A 28-year-old female diagnosed with PID after contracting chlamydia. Which of the following is a potential long-term consequence if treatment is delayed?

- (1) Increased fertility; no need for ART
- (2) Infertility risk; may need ZIFT or GIFT for conception
- (3) No infertility risk at all
- (4) Guaranteed spontaneous recovery without treatment

137. Consider the following statements about ART:

- A. IVF refers to fertilisation outside the body in almost similar conditions as that in the body.
- B. ZIFT is used when embryos have up to 32 blastomeres transferred into fallopian tube.
- C. GIFT involves transfer of unfertilized ovum into fallopian tube.
- D. ICSI is used when male partner has very low sperm count.
- E. AI cannot be used when female reproductive tract is completely blocked.

Which of the above statements is/are **correct**?

- (1) A, B and C only
- (2) A, B, C and D only
- (3) A, C, D and E only
- (4) A, B and D only

138. Match **List-I** with **List-II**.

List-I (Type of natural selection)		List-II (Characteristics)	
(A)	Directional selection	(I)	Selection favoring intermediate phenotypes
(B)	Stabilizing selection	(II)	Human breeding for specific traits
(C)	Disruptive selection	(III)	Selection pressure favours extreme phenotypes at both ends
(D)	Artificial selection	(IV)	Selection pressure favours one extreme phenotype

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-IV, B-I, C-III, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-IV, B-I, C-II, D-III

139. A couple seeks contraceptive advice. Female has severe nausea with oral pills, male has religious objections to sterilization. They want long-acting, reversible method with minimal side effects. Which contraceptive would be **most** appropriate?

- (1) Copper T (IUD)
- (2) Emergency contraceptive pills
- (3) Hormone releasing IUD
- (4) Vasectomy

140. Choose **correctly** paired match from the options given below.

- (1) Genetic drift - Acts stronger in large populations with gene flow
- (2) Mutation - Provides directional evolution without natural selection
- (3) Gene flow - Equalizes allele frequencies between populations
- (4) Natural selection - Random change in allele frequencies

141. Consider the following four statements (A-D) and select the **correct** option stating which ones are true (T) and which ones are false (F).

- A. Effective counselling on the need to avoid unprotected coitus can decrease the risk of an illegal medical termination of pregnancy.
- B. Contraceptives are regular requirements for the maintenance of reproductive health.
- C. In vasectomy, a small part of the vas deferens is removed or tied up through a small incision on the penis.
- D. Periodic abstinence method is highly effective but its reversibility is very poor.

- | | A | B | C | D |
|-----|----------|----------|----------|----------|
| (1) | F | F | T | T |
| (2) | T | F | F | T |
| (3) | F | T | T | F |
| (4) | T | F | F | F |

142. Given below are two statements:

Statement I: Louis Pasteur by careful experimentation demonstrated that life comes only from pre-existing life.

Statement II: Oparin of England and Haldane of Russia proposed that the first form of life could have come from pre-existing non-living organic molecules.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

143. Consider the following statements about evolution:
- Adaptations are heritable traits that increase organism's fitness.
 - All variations arise due to mutations.
 - Darwin's theory is based on observations of differential reproduction.
 - Lamarck proposed acquired characteristics could be inherited.
 - Genetic drift has no role in small populations.
- Which of the above statements is/are **correct**?
- A, C and D
 - A, B and C
 - B, D and E
 - A and C only
144. Which of the following **best** explains presence of gill slits in human embryos, though not functional in adults?
- They are acquired during foetal development due to maternal diet
 - They represent vestigial structures indicating common ancestry with aquatic vertebrates
 - They are functionless mutations with no evolutionary significance
 - They develop randomly and have no connection to evolution
145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Lactational amenorrhea method can be effective as a natural contraceptive up to a maximum period of 6 months following parturition only if the mother breast-feeds the child completely.
Reason R: Ovulation does not occur during the period of intense lactation following parturition.
 In the light of the above statements, choose the **correct** answer from the options given below.
- Both A and R are true, and R is the correct explanation of A.
 - Both A and R are true, but R is not the correct explanation of A.
 - A is true, but R is false.
 - A is false, but R is true.
146. Which of the following characteristic of population would **increase** effect of genetic drift?
- Large population size with random mating
 - Strong natural selection
 - High gene flow from neighboring populations
 - Small isolated population
147. Hormone-releasing IUDs prevents conception by which mechanism?
- Increasing phagocytosis of sperms only
 - Making uterus unsuitable for implantation and cervix hostile to sperms
 - Inhibiting ovulation completely
 - Killing sperms through toxic effects

148. Match **List-I** with **List-II**.

List-I		List-II	
(A)	IVF-ET	(I)	Embryo (up to 8 blastomeres) transferred into fallopian tube
(B)	ZIFT	(II)	The semen collected either from the husband or a healthy donor is artificially introduced either into the vagina or into the uterus
(C)	GIFT	(III)	Embryo transferred into uterus after <i>in vitro</i> fertilization
(D)	AI	(IV)	Unfertilized ovum transferred into fallopian tube

Choose the **correct** answer from the options given below:

- A-III, B-IV, C-I D-II
- A-IV, B-III, C-I, D-II
- A-III, B-I, C-IV, D-II
- A-III, B-I, C-II, D-IV

149. Hardy-Weinberg equilibrium will **not** be maintained if:

- There is random mating and no gene flow.
- Mutation rate is zero and population size is very large.
- There is natural selection favoring certain alleles.
- There are no geographic barriers between populations.

150. Consider statements about STI:

- Pelvic inflammatory disease (PID) results from untreated sexually transmitted infection and is a major cause of infertility.
- All STIs must be transmitted only through sexual contact.
- Early STI symptoms are minor - itching, discharge, slight pain.
- Infected females with STIs are always symptomatic.
- Hepatitis-B and HIV can be transmitted through blood transfusion and sharing needles.

Which of the above statements is/are **correct**?

- A, B and C
- A, C and E
- B, C and D
- A, D and E

151. In stable environment with no selection pressure, allele frequencies according to Hardy-Weinberg principle will:

- Increase progressively with each generation
- Decrease progressively with each generation
- Remain constant from generation to generation
- Fluctuate randomly

152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: According to the theory of natural selection, all variations in a population will be selected and passed on to the next generation.

Reason R: Natural selection acts on heritable variations that provide reproductive advantage in a given environment.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

153. Which of the following conditions regarding MTP in India is **correct** according to Amendment Act 2017?

- (1) MTP is completely banned under all circumstances
- (2) Pregnancy can be terminated up to 24 weeks with proper medical opinion and legal conditions
- (3) Amniocentesis for sex determination followed by MTP is legally permitted
- (4) MTP can be performed without medical practitioner's opinion

154. Concept that organisms with beneficial traits have increased chances of survival and reproduction is best described as:

- (1) Genetic drift
- (2) Natural selection
- (3) Mutation
- (4) Gene flow

155. Which of the following statements regarding early mammals and their evolutionary history are **incorrect**?

- A. The earliest mammals were small, shrew-like organisms with limited fossil evidence.
- B. Oviparity in mammals allowed better protection of developing young compared to reptiles.
- C. Early mammals outcompeted reptiles immediately after they first evolved.
- D. Continental drift allowed North American mammals to replace several South American mammalian forms after land connection formed.

E. Australian marsupials survived mainly due to the absence of competition from other mammals caused by geographic isolation.

- (1) D only
- (2) C and D only
- (3) B and C only
- (4) A and E only

156. In India, human population is heavily weighted towards the younger age groups as a result of:

- (1) Short life span of many individuals and high birth rate
- (2) Long life span of many individuals and high birth rate
- (3) Short life span of many individuals and low birth rate
- (4) Long life span of many individuals and low birth rate

157. Consider the following statements:

Statement I: Microbial experiments show that pre-existing advantageous mutations when selected will result in observation of new phenotypes.

Statement II: Hugo deVries based on his work on evening primrose brought forth the idea of mutations.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

158. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Mutation	(I)	Random change in allele frequency in small populations
(B)	Gene flow	(II)	Movement of genes between populations
(C)	Genetic drift	(III)	Source of new genetic variation
(D)	Natural selection	(IV)	Non-random survival and reproduction based on fitness

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-III, B-II, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-II, C-I, D-IV

159. Read the following statements regarding population control measures and the Reproductive and Child Health (RCH) programme in India:

- A. The statutory marriageable age was raised to 18 years for females and 21 years for males to check population growth.
- B. The popular slogan 'Hum Do Hamare Do' was introduced to promote a strict one-child norm.
- C. The RCH (Reproductive and Child Health) programmes cover a wider range of reproduction-related areas beyond just contraception.
- D. The slogan 'Hum Do Hamare Do' signifies a family size of a maximum of two children.
- E. Providing incentives to couples with small families was considered an ineffective measure and was discontinued.

Choose the **correct** option containing only the correct statements:

- (1) A, B and C (2) A, C and D
- (3) B, C and E (4) A, D and E

160. Match **List-I** with **List-II**.

List-I		List-II	
(A)	HIV	(I)	Caused by herpes simplex virus
(B)	Genital herpes	(II)	Caused by Hepatitis-B Virus
(C)	Hepatitis-B	(III)	Causes most fatal STI
(D)	Genital warts	(IV)	Caused by Human Papillomavirus (HPV)

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-III, C-I, D-IV
- (3) A-III, B-I, C-II, D-IV
- (4) A-III, B-I, C-IV, D-II

161. Given below are two statements:

Statement I: It is possible that the work of Thomas Malthus on natural selection influenced Darwin.

Statement II: A French naturalist Lamarck had said that evolution of life forms had occurred but driven by use and disuse of organs.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

162. Carefully observe the following diagram and choose **incorrect** statement regarding this.



- (1) It is made of thin rubber/ latex sheath.
- (2) It is used to cover the cervix in female.
- (3) It also protects the user from sexually transmitted disease.
- (4) It is reusable and can be self-inserted and thereby gives privacy to the user.

163. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Saheli	(I)	Barrier method
(B)	Vaults	(II)	Contraceptive implant
(C)	Multiload-375	(III)	Oral contraceptive pill
(D)	Norplant	(IV)	Intra uterine device

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-I, C-IV, D-III

164. A farmer continuously sprays a particular herbicide on his field for several years to control weeds. After some time, most weeds in the field become resistant to the same herbicide and continue to flourish, even though the herbicide has not been changed. Which of the following is the most appropriate biological inference from this observation?

- (1) Herbicide exposure induces specific mutations that direct weeds to become resistant over time.
- (2) Weeds were pre-adapted to the herbicide, so evolution was not involved in this change.
- (3) Randomly occurring resistant variants in the weed population were selected by anthropogenic action.
- (4) Evolution of herbicide resistance is a deterministic process with a pre-fixed goal in populations.

165. The main factor directly responsible for the explosive rise in human population during the last century is:

- (1) Rapid increase in birth rate and maternal mortality rate
- (2) Rapid decline in death rate, MMR and IMR with more people in reproductive age
- (3) Rapid decline in birth rate and increase in infant mortality rate
- (4) Constant death rate with rapidly falling birth rate

- 166.** Industrial melanism is an example of which type of adaptation or evolutionary phenomenon?"
- (1) Drug resistance
 - (2) Darkening of skin due to smoke from industries
 - (3) Protective resemblance with the surroundings
 - (4) Defensive adaptation of skin against ultraviolet radiation
- 167.** How many of the following are possible reasons for infertility in females?
- A. Physical
 - B. Congenital diseases
 - C. Drugs
 - D. Immunological reasons
 - E. Psychological reasons
- (1) Three only
 - (2) Two only
 - (3) Four only
 - (4) All five
- 168.** Which of the following options **correctly** represents the group of Australian marsupials showing adaptive radiation?
- (1) Lemur, Bandicoot and Wombat
 - (2) Marsupial mole, sugar glider and flying squirrel
 - (3) Bobcat, Koala and marsupial mole
 - (4) Tasmanian wolf, tiger cat and banded anteater
- 169.** Which of the following **correctly** represents the approximate change in population between 1900 and 2011?
- (1) World population: 2 billion to 7.2 billion; India: 350 million to 1.2 billion
 - (2) World population: 2 billion to 6 billion; India: 350 million to 7.2 billion
 - (3) World population: 6 billion to 7.2 billion; India: 1 billion to 2 billion
 - (4) World population: 2 billion to 7.2 billion; India: 1 billion to 350 million
- 170.** Which of the following statements about Sexually Transmitted Infections (STIs) is/are **correct**?
- A. HIV and hepatitis-B are the only STIs that are not completely curable.
 - B. Trichomoniasis, chlamydiasis, gonorrhoea, and syphilis can be fully cured if detected early.
 - C. Social stigma prevents many infected individuals from seeking timely diagnosis.
 - D. All STIs may lead to severe complications such as ectopic pregnancies, stillbirths, or reproductive cancers.
- E. Young individuals (15–24 years) show the highest reported incidence of STIs.
- (1) A and D only
 - (2) B, C and E only
 - (3) C, D and E only
 - (4) A, B, C and E
- 171.** Given below are two statements:
- Statement I:** Branching descent and natural selection are the two key concepts of Darwinian Theory of Evolution.
- Statement II:** According to Darwin, fitness is based on characteristics which are not inherited.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
- 172.** Which of the following statements is **not** correct regarding Hardy Weinberg principle?
- (1) The frequencies of AA, Aa and aa genotypes can be represented as p^2 , $2pq$ and q^2 respectively.
 - (2) The expression $p^2 + 2pq + q^2 = 1$ is a binomial expansion of $(p + q)^2$.
 - (3) Under genetic equilibrium, the sum of allelic frequencies satisfies $p + q = 1$.
 - (4) A deviation of observed genotype frequencies from $p^2 + 2pq + q^2$ indicates that the population is in equilibrium.
- 173.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion A:** Barrier contraceptive methods have very low failure rates when used properly and along with spermicidal creams.
- Reason R:** Condoms and diaphragms provide physical barriers that completely prevent sperm entry and also protect against STIs.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both A and R are true and R is the correct explanation of A.
 - (2) Both A and R are true but R is not the correct explanation of A.
 - (3) A is true but R is false.
 - (4) A is false but R is true.

174. What was the theory called that life could come from decaying matter like straw and mud?
- (1) Panspermia
 - (2) Chemical evolution
 - (3) Spontaneous generation
 - (4) Biogenesis

175. During which days of the typical 28 days menstrual cycle is the fertile period typically expected?
- (1) Day 1 to 7
 - (2) Day 10 to 17
 - (3) Day 18 to 28
 - (4) Day 14 to 21

176. Miller's experiment result in the formation of:
- (1) Living organisms
 - (2) Cells
 - (3) Amino acids
 - (4) Bacteria

177. Match List-I with List-II.

List-I		List-II	
(A)	Copper-T	(I)	Blocks gamete transport permanently through surgical intervention
(B)	Condoms	(II)	Increases phagocytosis of sperms and suppresses their motility
(C)	Oral contraceptive pills	(III)	Prevent conception by physical barrier
(D)	Vasectomy	(IV)	Inhibit ovulation and implantation; alter cervical mucus quality

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-III, C-I, D-IV
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-III, C-IV, D-I

178. Which of the following option contain the hormone-releasing IUDs only?

- (1) Lippes Loop, Multiload 375
- (2) Vaults, LNG-20
- (3) Multiload 375, Progestasert
- (4) Progestasert, LNG-20

179. Which of the following statements about the early universe and formation of earth is/are **correct**?

- A. The universe is estimated to be approximately 13.8 billion years old, according to the Big Bang theory.
 - B. Hydrogen and helium were among the first elements formed after the Big Bang as the temperature of the universe decreased.
 - C. Earth was formed around 4.5 million years ago.
 - D. Early Earth's atmosphere initially lacked gases such as methane, ammonia, and carbon dioxide.
 - E. Ozone layer formation began only after oxygen reacted with other early atmospheric gases.
- (1) A, B and E only
 - (2) A, B, C and E
 - (3) B, C and D only
 - (4) C, D and E only

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Antibiotic resistance in bacteria evolves very rapidly because bacteria reproduce quickly and have high mutation rates.

Reason R: Natural selection favors bacteria with resistance genes when antibiotics are present in the environment.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true, and R is the correct explanation of A.
- (2) Both A and R are true, but R is not the correct explanation of A.
- (3) A is true, but R is false.
- (4) A is false, but R is true.





PW Web/App - <https://smart.link/7wwosivoicgd4>

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Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____